Referee Report

Manuscript: *Numerical Spectral Correspondence between a Non-Autonomous Quadratic Map and the Riemann Zeros: An Exploratory Study*
Submitted to *Mathematical and Computational Applications*

Summary and general assessment

The manuscript explores a numerical correspondence between the imaginary parts of the non-trivial Riemann zeros and the eigenphases of an empirical transfer matrix constructed from the non-autonomous quadratic map

$$x_{n+1} = 1 - \lambda_n x_n^2, \qquad \lambda_n = \mu_c + \frac{k_1}{\log^2 n} + \frac{k_2}{\log^3 n}. \tag{1}$$

Long trajectories are used to build a finite-dimensional, real, non-normal and dissipative Markov matrix. After filtering, ordering, phase unwrapping and linear rescaling, its eigenphases are compared with known Riemann zeros. The author studies the first few zeros, a residual feature near $N \simeq 20$, larger-scale fits, conjugate-symmetry effects, out-of-sample validation and unfolded spacing statistics.

I found the paper original and interesting. Its bottom-up dynamical approach is quite different from the usual Hilbert–Pólya or Berry–Keating programs, and I appreciate the unusually explicit account of the tests that do not support the stronger interpretation of the model. Table 2 is particularly useful in this respect. In my view, however, the manuscript is much too long, and the scientific message is obscured by repeated technical descriptions and by claims that are sometimes stronger than the numerical evidence. I would be willing to recommend publication after a major revision along the following lines.

Main comments

1. **What is fitted, and what is predicted?**
   My main concern is that several ingredients—the baseline, logarithmic couplings, discretization scale, spectral cutoff and final linear rescaling—are calibrated using the same zeros against which the spectrum is evaluated. The low-order agreement is therefore an *in-sample fitted correspondence*, rather than an independent generation of the zeros.
   I find the out-of-sample test particularly revealing. The test MSE is one to two orders of magnitude larger than the training MSE, while the fitted k_1 changes substantially with the calibration window. This seems to show that the present construction does not predict unseen zeros. I suggest making this one of the principal conclusions of the paper, rather than treating it mainly as a technical limitation.
   The cooling law also needs a sharper presentation. The $1/\log^2 n$ term results from combining the known mean spacing of the zeros with an assumed square-root response of the eigenphases; it is not derived from the spectral theory of the map. Similarly, the $1/\log^3 n$ term appears to be an additional fitting correction. The observed drift of k_1 with fitting range and grid resolution should not be interpreted as an RG flow unless a beta function, fixed point or universality statement can be established.
2. **What operator is actually being diagonalized?**
   For a non-autonomous system, the evolution over T steps is described by

   $$P_{T-1}P_{T-2}\cdots P_0, \tag{2}$$

   whereas the paper diagonalizes a time-averaged empirical transition matrix. In general, the two spectra need not be related. Can the author compare these objects in a small system, or give an argument showing what information survives the averaging? If not, the paper should state very clearly that the eigenvalues of the averaged matrix have not been shown to be resonances of the original non-autonomous dynamics.

There is a related numerical issue. The final sequence depends on smoothing or Monte Carlo reconstruction, grid resolution, an eigenvalue cutoff, selection of one half-plane, sorting, unwrapping and regression. Sorting and unwrapping impose a linear order on an unordered complex spectrum. A short sensitivity table would help establish which features are robust.

I was also concerned that the microscopic Gaussian-smoothed construction and the macroscopic Monte Carlo construction do not yield consistent eigenphases in their overlap regime. The large-N calculation should therefore be described as a related but distinct numerical construction, not simply as a scalable continuation of the microscopic one.

3. **GUE and the meaning of the large-N agreement.**
The globally rescaled GUE comparison concerns the mean counting function, whereas the Montgomery–Odlyzko correspondence concerns unfolded local statistics. Thus, the poorer global GUE fit is neither evidence against random-matrix theory nor evidence that the proposed dynamics gives a more fundamental description.
The unfolded calculation in the manuscript is, to me, the relevant test: the Riemann zeros are compatible with GUE, while the model eigenphases are incompatible with GUE and compatible with Poisson statistics. Hence the model may reproduce part of the smooth mean trend after calibration, but it does not reproduce the local level repulsion characteristic of the zeros. This should be stated directly in the Conclusions.
The large-N sections also need a more modest interpretation. Since the parameters are optimized against the target zeros, the calculations mainly show that the imposed logarithmic schedule can be fitted to their mean density over a large range. They do not establish a predictive spectral correspondence.

4. **Ion-trap comparison and relation to earlier work.**
Every substantive discussion of the ion-trap realization—the protocol, measured zeros, error bars, operating parameters and possible decoherence mechanisms—should cite Refs. [26] and [27], rather than citing them only at the first mention. The Spearman correlation is worth reporting, but the specific co-location near $N \simeq 20$ has $p = 0.099$ and is not statistically significant at the usual level. Moreover, the numerical residual and experimental uncertainty arise from different mechanisms. I would therefore shorten the speculation about a logarithmic-manifold obstruction and state plainly that the present model does not explain the ion-trap data. Nor has the proposed *single-sided truncation* been shown to represent the experimental Hamiltonian or measurement protocol.
The construction is motivated by the author's earlier Ref. [1], where the symbolic dynamics at the logistic-map band-merging point is related to the prime sieve. Since this result is essential motivation for the present work, I suggest adding a short appendix summarizing the precise result of Ref. [1], the assumptions under which the claimed isomorphism holds, and exactly which part of that structure is retained in the present non-autonomous map. This would make the paper more self-contained without overloading the Introduction.
I would also like to ask whether the author has considered applying the same procedure to zeros of Dirichlet L-functions. This would provide a useful test of whether the construction captures arithmetic information beyond the mean zero density. In particular, can one modify the dynamics so that it distinguishes different characters or symmetry classes, rather than merely fitting a generic logarithmic counting law? Even a brief discussion of this question would improve the outlook section.

5. **Reproducibility, scope and presentation.**
Nominally identical unseeded optimizations produce values such as $k_1 \simeq 6.48$ and 8.07. The main calculations should use fixed reported seeds, quote distributions rather than a preferred optimum, and identify clearly which run is used in each figure. No single fitted value appears canonical at present.
The distinction from Hilbert–Pólya should also remain explicit. The empirical matrix is real, non-symmetric, non-normal and dissipative, and the quantities compared with the zeros are processed arguments of complex eigenvalues. The paper does not construct a self-adjoint operator, a spectral determinant for $\zeta(s)$, a prime-orbit trace formula, or an argument bearing on the Riemann Hypothesis.
Finally, I strongly recommend shortening the manuscript. The main text could focus on the map, the empirical matrix, the low-order fit, out-of-sample validation, the unfolded-spacing test and a concise discussion. Detailed parameter scans, Gaussian splatting, Monte Carlo implementation, eigensolvers, optimizer settings, individual seeds, extended ablations, long tables and stress tests should be moved to appendices or supplementary material. Terms such as *phase transition*, *topological mutation*, *intrinsic physical noise floor* and *rigorous higher-order perturbative form* should be replaced by more precise numerical language. I would also replace *ground-state zero* by *first non-trivial zero* or *lowest positive ordinate.*

Conclusions and recommendation

The Conclusions should be shorter and separate positive observations from limitations. In my view, the main message is the following:

> The proposed system gives a phenomenological, calibrated reconstruction of part of the smooth counting-function behavior of the Riemann zeros, but it does not predict unseen zeros, reproduce their GUE local statistics, or furnish a self-adjoint spectral realization of the zeta function.

The manuscript contains an original and potentially useful exploratory idea, and its transparent presentation of failed extrapolation and non-GUE statistics is a real strength. Nevertheless, the present version is too long and occasionally overinterprets a calibrated numerical fit. I recommend **major revision**. A shorter and clearer account of what the model does and does not achieve could make an interesting contribution to *Mathematical and Computational Applications*.